

Jason Wakim Richards

+44 77 08 348 077 | jason2richards@gmail.com

github.com/richarj2002 | linkedin.com/in/jason-richards-jwr2002/ | richarj2002.github.io/personal_website/

Guidance, navigation, and control engineer with hands-on experience developing, integrating, and testing autonomy for planetary rovers, spacecraft, and UAVs. At Airbus Defence & Space, I led LiDAR test activities for an autonomous Sample Fetch Rover concept, built a networked sensor test bench, characterized sensor performance, and developed C/C++ simulation software that generated synthetic imagery and LiDAR scans. I now develop and flight-test UAV localization, mapping, planning, and control using ROS 2, PX4, Linux, Python, C++, cameras, and IMUs. As ADCS Lead for the JamSail 3U CubeSat, I defined requirements and interfaces, developed embedded flight software and simulation, and implemented an EKF for multi-sensor attitude estimation. I bring practical experience across GNC requirements, algorithm development, sensor calibration, performance analysis, hardware/software integration, and verification and validation.

Work Experience

Doctoral Researcher – Autonomous UAV Systems

November 2025 – Present

SnT, University of Luxembourg | Kirchberg, Luxembourg

- Develop C++ and Python localization, Semantic Relational SLAM, and situational-awareness algorithms for UAVs, linking camera/IMU perception and state estimation with planning and control.
- Integrate, tune, and deploy autonomous navigation on physical UAVs using ROS 2, PX4, embedded Linux, Raspberry Pi, and OpenVINS visual-inertial odometry.
- Build reproducible Docker-based development and deployment environments and camera-IMU characterization and calibration workflows with Kalibr, supported by version-controlled software and technical documentation.
- Plan and conduct simulation, laboratory, hardware, and flight testing; analyze integrated performance and troubleshoot interfaces across sensing, estimation, mapping, planning, and control.
- Collaborate across perception, autonomy, and vehicle-integration activities while documenting algorithms, interfaces, test results, and technically grounded next steps.

AOCS/GNC Engineer Intern

July 2023 – August 2024

Airbus Defence & Space | Stevenage, UK

- Served as lead test engineer for LiDAR activities during Integrated BreadBoarding (iBB) 3 & 4, planning procedures and supporting end-to-end verification of the Sample Fetch Rover concept at Technology Readiness Level 6.
- Designed and built an integrated hardware/software GNC test bench, connecting LiDAR to rover breadboards through ROS over a local wireless network to capture data, characterize sensor performance, and validate navigation models.
- Developed C/C++ software on Linux for an offline GNC simulation, coupling a natural-scene generator to produce synthetic camera imagery and LiDAR scans for rover algorithm development and performance analysis.
- Investigated LiDAR and radar simulation, terrain generation, and Vector Field Histogram planning for lunar-rover navigation, translating prototype concepts into the existing simulation environment.
- Analyzed test-campaign data and documented results while working across GNC, simulation, software, and test disciplines using Git, GitLab, Jira, and Confluence in a fast-paced Agile team.

Projects & Leadership

JamSail 3U CubeSat – AOCS/ADCS Lead

September 2024 – August 2025

NottSpace, University of Nottingham | Nottingham, UK

- Led ADCS design and implementation for a NottSpace 3U CubeSat, defining subsystem and algorithm requirements, concept of operations, architecture, interfaces, and a verification and validation plan.
- Developed embedded C flight software and a C/C++/CMake GNC simulation on Linux to design and verify attitude determination and control algorithms before deployment to flight hardware.
- Designed and implemented an EKF that fuses sun-sensor, magnetometer, and IMU measurements, together with TRIAD attitude determination and control-state logic.
- Characterized sensor noise with FFT analysis, tuned digital low-pass filters, and verified their performance through simulation and hardware testing; implemented the IGRF magnetic-field model as a physics-based simulation input.

Nottingham Space Society | Nottingham, UK

- Defined system requirements, architecture, subsystem interfaces, Project Breakdown Tree, and Work Breakdown Structure within a student-led, 12-person team developing a 3U CubeSat.
- Developed software on a Raspberry Pi running Linux and integrated a microcontroller, IMU, and cameras over I2C before the concept progressed to the competition build phase.
- Contributed across embedded software, hardware and sensor integration, and system simulation while coordinating interfaces and technical development within the student team.
- Achieved a top-four finish among 12 UK teams as first-time entrants to the UKSEDS Satellite Design Competition.
- Led cross-team meetings, communicated design changes, and presented the integrated concept to industry experts, balancing technical interfaces, build constraints, and competition deadlines.

UKSEDS Volunteer

Sept. 2024 – Sept. 2026

UK Students for the Exploration and Development of Space | Remote & UK field events

- Served as an Olympus Rover Trials (ORT) volunteer in the first year, helping organise a UK national rover competition with Airbus Defence & Space and RAL Space.
- Progressed in the second year to Competitions Management Co-Lead, overseeing four national competitions: ORT, SDC, ISAM, and NRC.
- Coordinated volunteers, schedules, technical milestones, participant support, and partner engagement with Airbus Defence & Space (SDC), the UK Space Agency (ISAM), Midlands Rocketry Club, and Skyrora (NRC).
- Produced clear participant documentation and supported venue and field-event logistics, helping multidisciplinary volunteer teams deliver competitions to fixed schedules.

Education & Training**MEng Aerospace Engineering – First-Class Honours**

September 2020 – August 2025

University of Nottingham | Nottingham, UK

- Graduated with First-Class Honours, specializing in Aerospace Control Systems, Avionics, dynamics, software, modeling, and simulation.
- Applied C/C++, Python, MATLAB, Simulink, and Linux to estimation, control, flight dynamics, and autonomous-system projects.
- Completed design projects spanning aerospace systems, avionics, simulation, and control, providing the systems-engineering foundation for later space and robotics work.

IEEE RAS Summer School on Multi-Robot Systems

July 2026

Czech Technical University in Prague | Prague, Czechia

- Completed the 2026 IEEE RAS Summer School on Multi-Robot Systems, covering cooperative aerial robotics and active perception; presented the SaCoGraph collaborative-UAV research concept.
- Applied multi-robot autonomy concepts to collaborative sensing, semantic mapping, and resource-aware mission planning.

Space Resources & Robotics Summer School

May 2026

AGH University of Kraków | Krakow, Poland

- Completed intensive training in space resources, planetary robotics, and robotic manipulation for exploration applications.
- Developed a broader systems perspective on robotic operations in challenging off-world environments.

Space Systems Engineering Training Course

May 2025

ESA Academy | Transinne, Belgium

- Completed intensive ESA Academy training across the space-system lifecycle, from mission objectives and concepts of operations through architecture, integration, verification, and operations.
- Applied requirements engineering and concurrent-design methods to translate user needs into traceable, testable requirements and manage interfaces between technical disciplines.
- Practised system options and trade studies, budgets and margins, verification and validation planning, and the relationship between engineering decisions, schedule, cost, and risk.
- Completed multidisciplinary group exercises and communicated design rationale and systems-engineering deliverables, strengthening collaborative problem-solving across complex spacecraft projects.